

# R Code and Output for Global Insect Richness

Load libraries and ../Data

```
library(SpadeR)
library(dplyr)
library(stringr)
library(pairwiseCI)
#library(remotes)
#install_version("CARE1", "1.1.0")
library(CARE1)

## Load Data

#Microgastrinae

reared <- read.csv("../Data/Microgastrinae_ACG_reared.csv") %>%
  select(BIN, Frequency)

core <- read.csv("../Data/Microgastrinae_ACG_Core_Malaise.csv") %>%
  select(BIN, Frequency)

peri <- read.csv("../Data/Microgastrinae_ACG_Peripheral_Malaise.csv") %>%
  select(BIN, Frequency)

# All insecta core

insc <- read.csv("../Data/Insecta_ACG_Core_Malaise.csv") %>%
  select(BIN, Frequency)

## highlight estimator

estimator <- "      iChao1 (Chiu et al. 2014)"
```

```

## adding in the trees

#TA = number of tree species in ACG
TA <- 1500
TA_alt <- 1200

#Tc = number of tree species in CR

TC <- 2500
#Tw = number of tree species on Earth

## estimated tree species

TW <- 73274
TW_CI <- c(73271, 73276)

TW_2 <- 89147

## described number of tree species

TW_d <- 60065

### Number of Saturniid moths

SA <- 106
SW <- 4535

```

## Approach 1: Calculate insect richness worldwide using the CARE estimator for Microgastrinae

```

### Total number of BINS

reared %>% nrow()

```

```
[1] 889
```

```
core %>% nrow()
```

```
[1] 388
```

```
peri %>% nrow()
```

```
[1] 576
```

```
###
```

### Step 1: Combining Malaise and Reared samples of Microgastrinae and estimating the true richness of ACG Microgastrinae

```
full_micro_table <- reared %>%  
  rename(reared = Frequency) %>%  
  full_join(rename(core, core = Frequency)) %>%  
  full_join(rename(peri, peri = Frequency)) %>%  
  mutate(reared = ifelse(is.na(reared), 0, 1)) %>%  
  mutate(core = ifelse(is.na(core), 0, 1)) %>%  
  mutate(peri = ifelse(is.na(peri), 0, 1))
```

Joining with `by = join\_by(BIN)`

Joining with `by = join\_by(BIN)`

```
head(full_micro_table)
```

|   | BIN          | reared | core | peri |
|---|--------------|--------|------|------|
| 1 | BOLD:AAA1266 | 1      | 0    | 0    |
| 2 | BOLD:AAA1634 | 1      | 1    | 1    |
| 3 | BOLD:AAA2143 | 1      | 0    | 0    |
| 4 | BOLD:AAA2511 | 1      | 0    | 0    |
| 5 | BOLD:AAA2668 | 1      | 0    | 0    |
| 6 | BOLD:AAA3229 | 1      | 0    | 1    |

```
nrow(full_micro_table)
```

```
[1] 1414
```

```
# Transform observed ../Data to CARE1 input format  
micro_care <- as.record(full_micro_table[, -1])  
  
micro_care
```

```
001 010 011 100 101 110 111
246 132 147 666 114 40 69
```

```
set.seed(100)

# Main step to obtain population size estimates
CARE1.print(micro_care)
```

Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.  
Use c() or as.vector() instead.

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(1) NUMBER OF IDENTIFIED CASES IN EACH LIST:

```
n1  n2  n3
889 388 576
```

(2) ESTIMATES BASED ON ANY PAIR OF SAMPLES:

|      | Petersen | Chapman | se  | cil  | ciu  |
|------|----------|---------|-----|------|------|
| pa12 | 3165     | 3146    | 237 | 2734 | 3667 |
| pa13 | 2798     | 2790    | 151 | 2522 | 3116 |
| pa23 | 1035     | 1033    | 37  | 970  | 1115 |

Note1: Refer to Seber(1982,pages 59 and 60) for Petersen estimator and Chapman estimators as well as s.e formula.

Note2: A log-transformation is used is used to obtain the confidence interval so that the lower limit is always greater than the number of ascertained. Refer to Chao(1987,Biometrics,43,783-791) for the construction of the confidence interval.

(3) SAMPLE COVERAGE APPROACH:

|        | M    | D    | Chat  | est  | se  | cil  | ciu  |
|--------|------|------|-------|------|-----|------|------|
| Nhat-0 | 1414 | 1066 | 0.495 | 2156 | 66  | 2038 | 2296 |
| Nhat   | 1414 | 1066 | 0.495 | 3405 | 638 | 2493 | 5088 |
| Nhat-1 | 1414 | 1066 | 0.495 | 2394 | 97  | 2221 | 2604 |

Warning: The estimated sample coverage(overlapping information)is too low so that Nhat is unstable. Recommend the use of Nhat-0 or Nhat-1.  
Parameter estimates:

|        | u1   | u2   | u3   | r12   | r13   | r23  |
|--------|------|------|------|-------|-------|------|
| Nhat-0 | 0.41 | 0.18 | 0.27 | -0.32 | -0.23 | 1.08 |
| Nhat   | 0.26 | 0.11 | 0.17 | 0.08  | 0.22  | 2.29 |
| Nhat-1 | 0.37 | 0.16 | 0.24 | -0.24 | -0.14 | 1.31 |

Definitions for the sample coverage approach:

M: number of individuals ascertained in at least one list.

D: the average of the number of individuals listed in the combination of any two lists omitting the other one.

$C^{\sim}$ : sample coverage estimate, see Equation (14) of Chao and Tsay(1998).  
est: population size estimate.

se: estimated standard error of the population size estimation based on bootstrap replications.

cil: 95% confidence interval lower limit(using a log-transformation).

ciu: 95% confidence interval upper limit(using a log-transformation).

Nhat: Population size estimate for sufficiently high sample coverage cases, see Equation (20) of Chao and Tsay (1998).

Nhat-1: One-step population size estimate for low sample coverage cases; see Equation (2.21) of Chao et al. (1996). This estimator is suggested for use when the estimated se of Nhat is relatively large.

u1,u2,u3: estimated mean probabilities depending on the estimate of N.

r12,r13,r23 etc.: estimated coefficient of covariation(CCV) depending on the estimate of N.

```
# Print the CARE1 input format

## n1 = reared = 886
## n2 = core = 388
## n3 = peripheral = 576

## Using Nhat-1

Nhat1 <- round(estN(micro_care, method = "LSC", se = TRUE, nboot = 200))

micro_all_estimate <- Nhat1[1]
micro_all_CI <- c(Nhat1[3], Nhat1[4])
```

## Step 2. Estimating the undersampling ratio for Core Microgastrinae.

```
## proportion of core malaise of the ichao total  
proportion_core <- nrow(core)/micro_all_estimate  
proportion_core
```

```
[1] 0.1620718
```

```
CI_proportion_core <- nrow(core)/micro_all_CI  
CI_proportion_core
```

```
[1] 0.1758043 0.1478095
```

## Step 3 Calculate total number of insects in ACG

```
## Total number of insecta bins in ACG core malaise  
n_insects_core <- nrow(incsc)  
n_insects_core
```

```
[1] 53945
```

```
total_insec_acg <- n_insects_core*(1/proportion_core)  
total_insec_acg
```

```
[1] 332846.2
```

```
CI_total_insect_acg <- n_insects_core*(1/CI_proportion_core)  
n_insects_core*(1/CI_proportion_core)
```

```
[1] 306846.9 364963.0
```

```
n_insects_core*(1/CI_proportion_core)
```

```
[1] 306846.9 364963.0
```

```
CI_total_insect_acg
```

```
[1] 306846.9 364963.0
```

**Step 4. Estimating global insect richness — See Appendix S3 (Excel spreadsheet)**